INCH-POUND

A-A-59753
25 October 2004

COMMERCIAL ITEM DESCRIPTION

GENERATOR, FUNCTION
(0.1 Hz to 4 MHz)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes the minimum requirements for a Function Generator. The equipment, as described in this CID, shall have a minimum frequency range of 0.1 Hz to 4 MHz. This CID is meant as a minimum requirement for the function generator in which only those manufacturers that meet or surpass the following requirements are to supply this generator per this CID.

2. Classification. Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet the following description. The equipment shall be Class 3, in accordance with MIL-PRF-28800, except as specified herein.

3. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

3.1 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein except as specified herein.

3.1.1 Temperature Operating. The equipment shall conform to its specified performance and accuracy while being operated in a temperature range of 0°C to +50°C.

3.1.2 Temperature Non-operating. The equipment shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -20°C to +70°C.

3.2 Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 120 VAC (±10 %) at line frequencies of 48 Hz to 63 Hz and 240 VAC (±10 %) at line frequencies of 48 Hz to 63 Hz. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein.

3.3 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 4000 hours of operation with a statistical certainty of 95%.

3.4 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. The calibration adjustments, wherever possible, shall be accessible without removal of the equipment case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours. No special tools shall be required for alignment or adjustment except those furnished with the equipment.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:26Z
Check the source to verify that this is the current version before use.

3.5 Performance Requirements. The equipment shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions and operating service condition as specified in MIL-PRF-28800.

3.5.1 Frequency Range. The frequency range shall be at least 0.1 Hz to 4 MHz.

3.5.1.1 Accuracy. The frequency accuracy shall be at least ± 5 % of full scale.

3.5.2 Waveforms. The function generator shall have as a minimum a sine wave, triangle wave, square wave and pulse output.

3.5.2.1 Sine Wave Distortion. The sine wave distortion shall not be greater than ± 1% to 1 MHz. From 1 MHz to 4 MHz the distortion shall be at least 25 dB below the fundamental.

3.5.2.2 Triangle Linearity. The triangle linearity shall be at least 95% or better to 200 kHz.

3.5.2.3 Square wave and Pulse Rise and Fall Time. The square wave and pulse rise and fall time shall be less than 50ns for 10 V_{pp} into a 50 ohm load.

3.5.3 Operational Modes. The function generator shall have at a minimum the following selectable modes:

3.5.3.1 Triggered Mode. Manual or external trigger initiates one complete waveform. The trigger shall at least meet the following characteristics:

Impedance: 10 k Ω ± 5%
Maximum Input Voltage: TTL compatible
Minimum Pulse Width: 100 ns

3.5.3.2 Gated Mode. Manual or external trigger initiates a continuous wavetrain while the trigger signal is applied. The gated mode trigger shall at least be the same as the triggered mode characteristics.

3.5.3.3 Continuous Mode. Generator provides an uninterrupted wave train at the selected frequency.

3.6.4 Sweep Mode. The function generator shall have a linear and a logarithmic sweep mode. The sweep shall include a selectable sweep range of 1:1000 in the linear mode and 1:100000 in the logarithmic mode.

3.7 Physical Characteristics. The function generator's component dimensions and weight shall be consistent with current commercial offerings. The equipment shall have the following physical characteristics.

3.7.1 Dimensions. The width and depth of the equipment shall be compatible with mounting in an ANSI/EIA-310-D standard equipment rack. The height shall be compatible with four standard height units, 4U. One unit or "U" is defined by ANSI/EIA-310 to be 1.75 inches or 44.45 mm.

3.7.2 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed that of a one-man lift, 37 lbs (or 16.78 kg). If the depth of the equipment is greater than 24 inches the maximum allowable weight shall be reduced by 33 percent.

3.7.3 Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

3.7.4 Signal Input and Output Connectors. Unless otherwise specified, all signal input and output connectors shall be Type BNC female.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:26Z
Check the source to verify that this is the current version before use.

3.7.5 Manuals. The equipment shall be delivered with operator, maintenance, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

4. REGULATORY REQUIREMENTS.

4.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5.3 The contracting officer has the option of accepting or rejecting the product.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Sources of documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

7.1.3 IEEE standards can be obtained from IEEE Customer Service Center, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331, U.S.A. Phone: , 1-800-678-4333 (Toll-Free, USA & Canada), 1-732-981-0060 (Worldwide), Fax:, 1-732-981-9667. Website: http://www.ieee.org.

7.1.4 ANSI/EIA standards can be obtained from the Electronics Industry Association, Engineering Department, 2001 Pennsylvania Ave., N.W., Washington, D.C., 20006. Phone: 1-800-854-7179 (USA and Canada).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:26Z
Check the source to verify that this is the current version before use.

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

Agent:
Air Force - 99

(Project 6625-F097)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://dodssp.daps.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:26Z
Check the source to verify that this is the current version before use.